

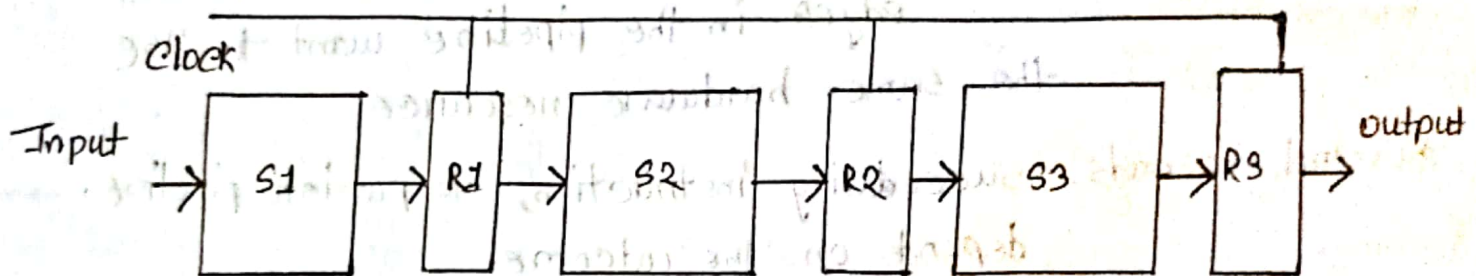
Part-B

Segment - 6 (Pipeline Hazard)

Pipelining: pipelining is a technique where multiple instructions are overlapped during execution.

- ⇒ Does not reduce a single task.
- ⇒ Pipeline rate limited by longest stage.

R - Register, S - Stage



Advantage: ① The cycle time of the processor is reduced.

② It increases the throughput of the system.

③ It makes the system reliable.

④ In every clock cycle, a new instruction finishes its execution.

⑤ Allow multiple instruction to be executed concurrently.

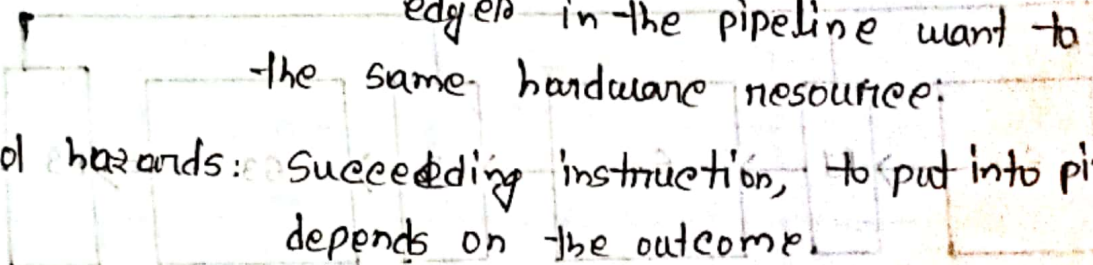
Pipelining MIPS:

What makes it easy with MIPS / Advantages:

- All instructions are same length.
- Just a few instruction format
- Memory operands appear only in load/stores
- operands are aligned in memory.

What makes it hard? / Disadvantages:

- Structural hazards: Different instructions, at different edges in the pipeline want to use the same hardware resource:



- Control hazards: Succeeding instruction, to put into pipeline, depends on the outcome.

- Data hazards: An instruction in the pipeline requires data to be computed by a previous instruction.

Pipeline Stages of MIPS instruction: There are 5 steps in instruction execution.

- ① Instruction Fetch and PC Increment (IF)
- ② Instruction Decode and Register Read (ID)
- ③ Execution or calculate address (EX)
- ④ Memory access (MEM)
- ⑤ Write result into register (WB)

Pipeline hazard: Pipeline hazards are situations that prevent the CPU from executing instructions smoothly and efficiently in a pipelined architecture. Pipelining is a technique that divides an instruction into smaller stages and allows multiple instructions to be

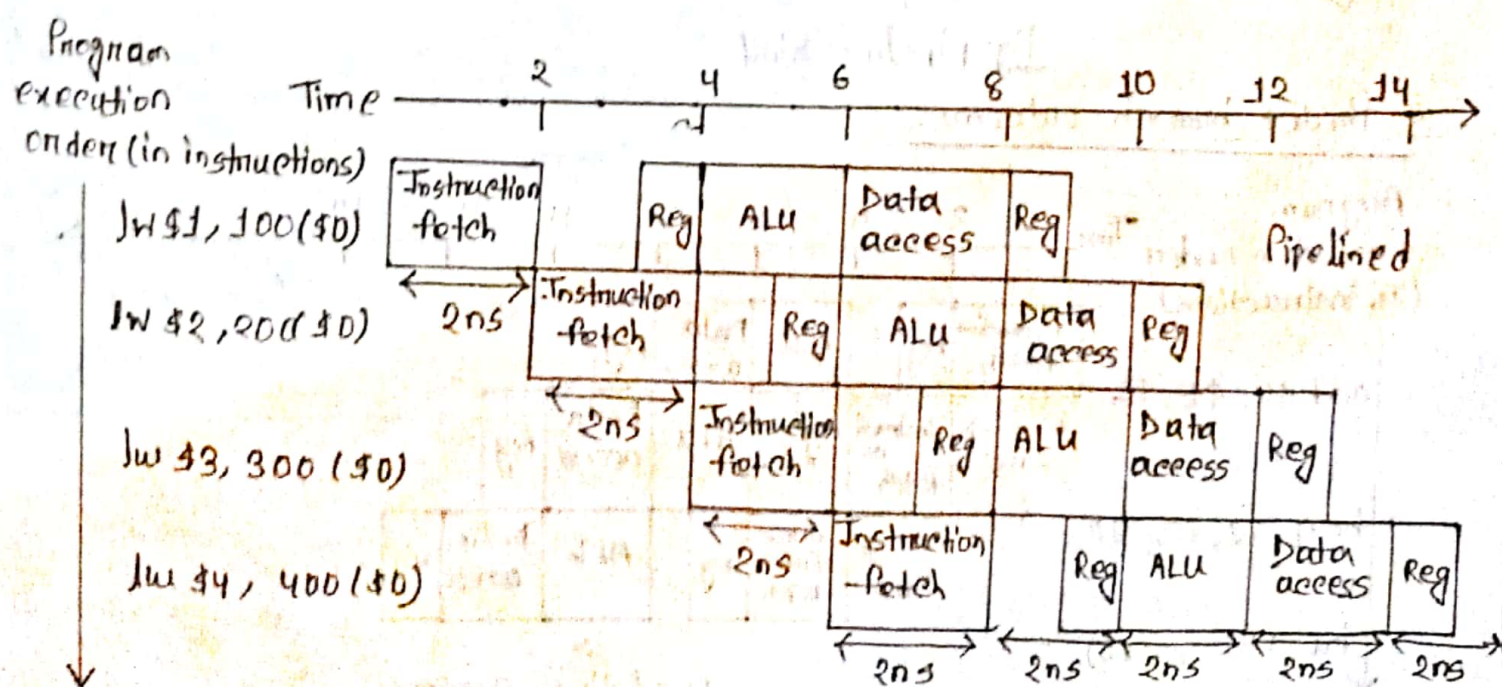
There are some kinds of situations when the next instruction cannot be executed in the following clock cycle in pipelining implementation. These events are called hazards.

There are three types of hazards. They are, ① Structural hazard
② Data hazard
③ Control hazard

Structural hazard: An instruction cannot be executed because the hardware cannot support the combination of instructions that are set to execute in the given clock cycle.

⇒ Inadequate hardware to simultaneously support all instructions.

⇒ Structural hazard can be avoided by stalling or duplicating the resource or pipelining the resource.



Control hazard: An instruction cannot be executed because the instruction that was fetched is not the one that is needed. This is Control hazard.

⇒ It need to make a decision based on the result of a previous instruction still executing in pipeline.
 ⇒ It can be avoided by insert a pipeline we use prediction.

Solution: ① Stall the pipeline ② Predict branch outcome

① Stall the pipeline:

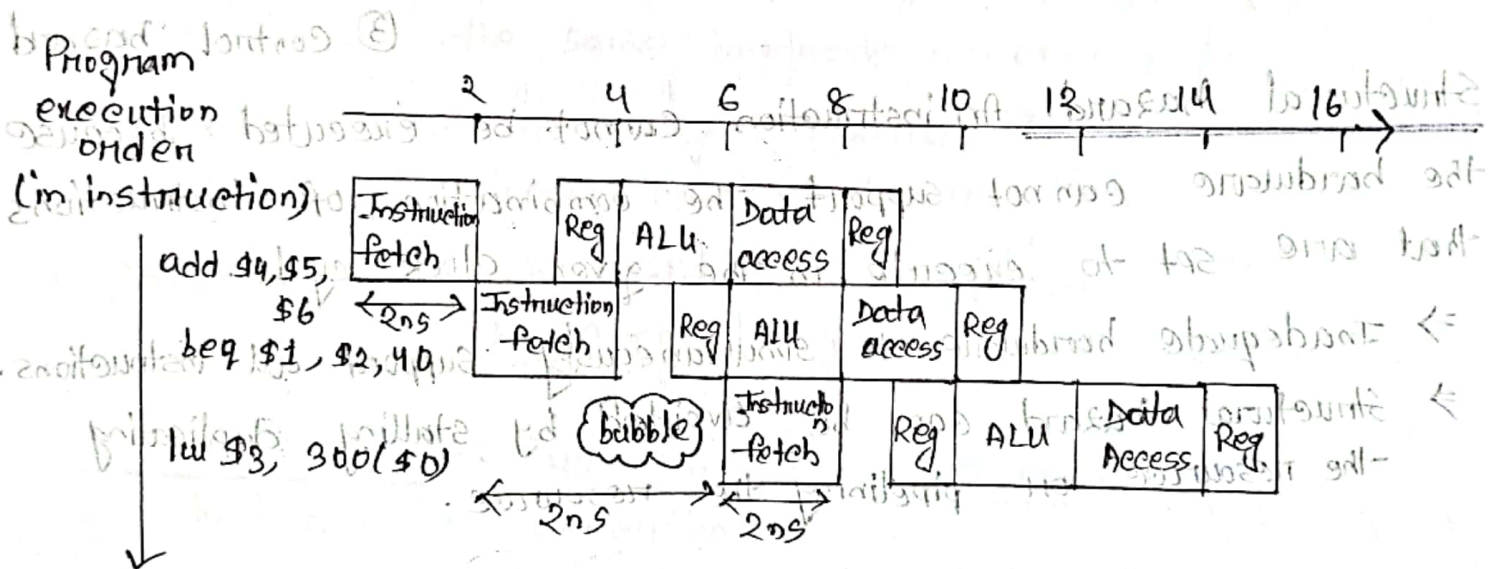
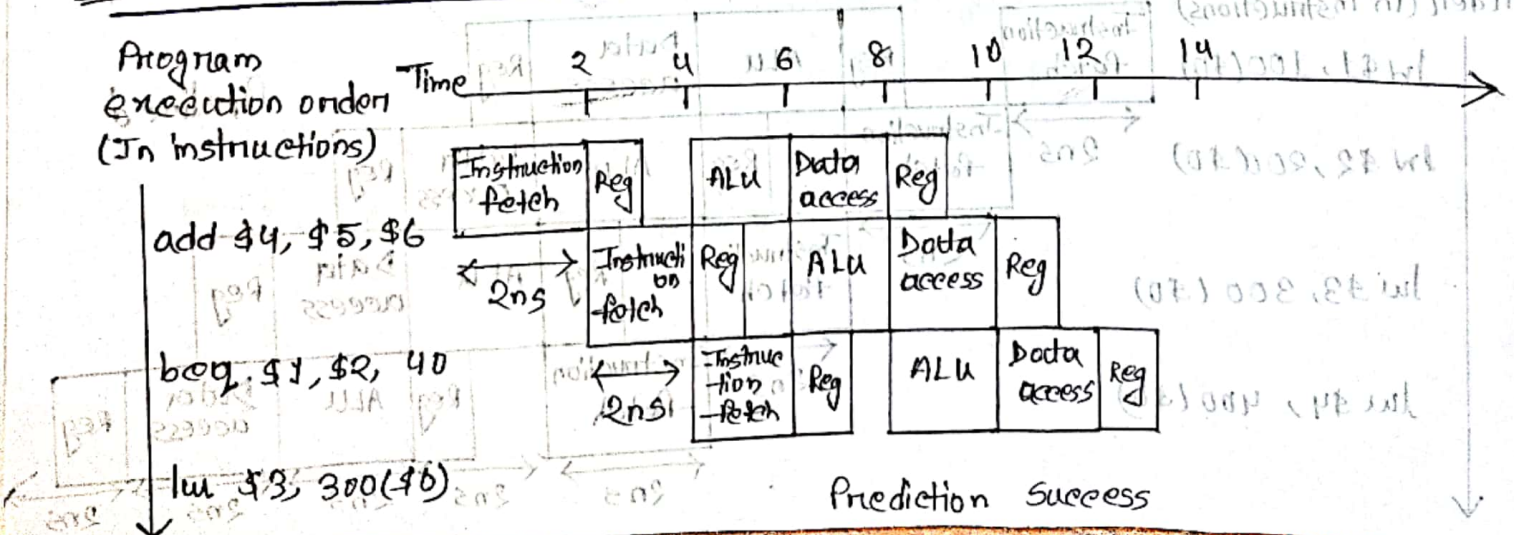
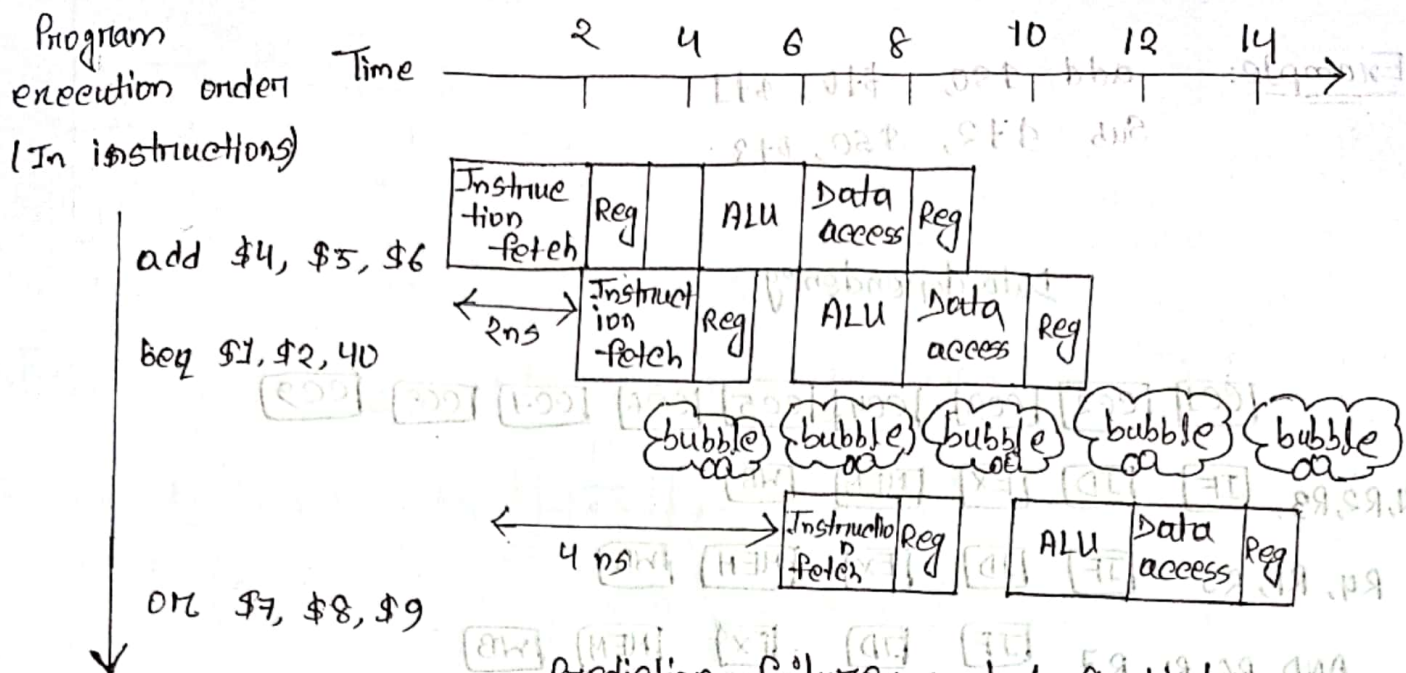


Fig: Pipeline stall

② Predict branch outcome:





Data Hazard: Data hazard also called pipeline data hazard. A planned instruction cannot execute in the proper clock cycle because data that is needed to execute the instruction is not yet available.

⇒ Data hazard occurs when two instructions in a pipeline refer to the same register and at least one of them writes to the register.

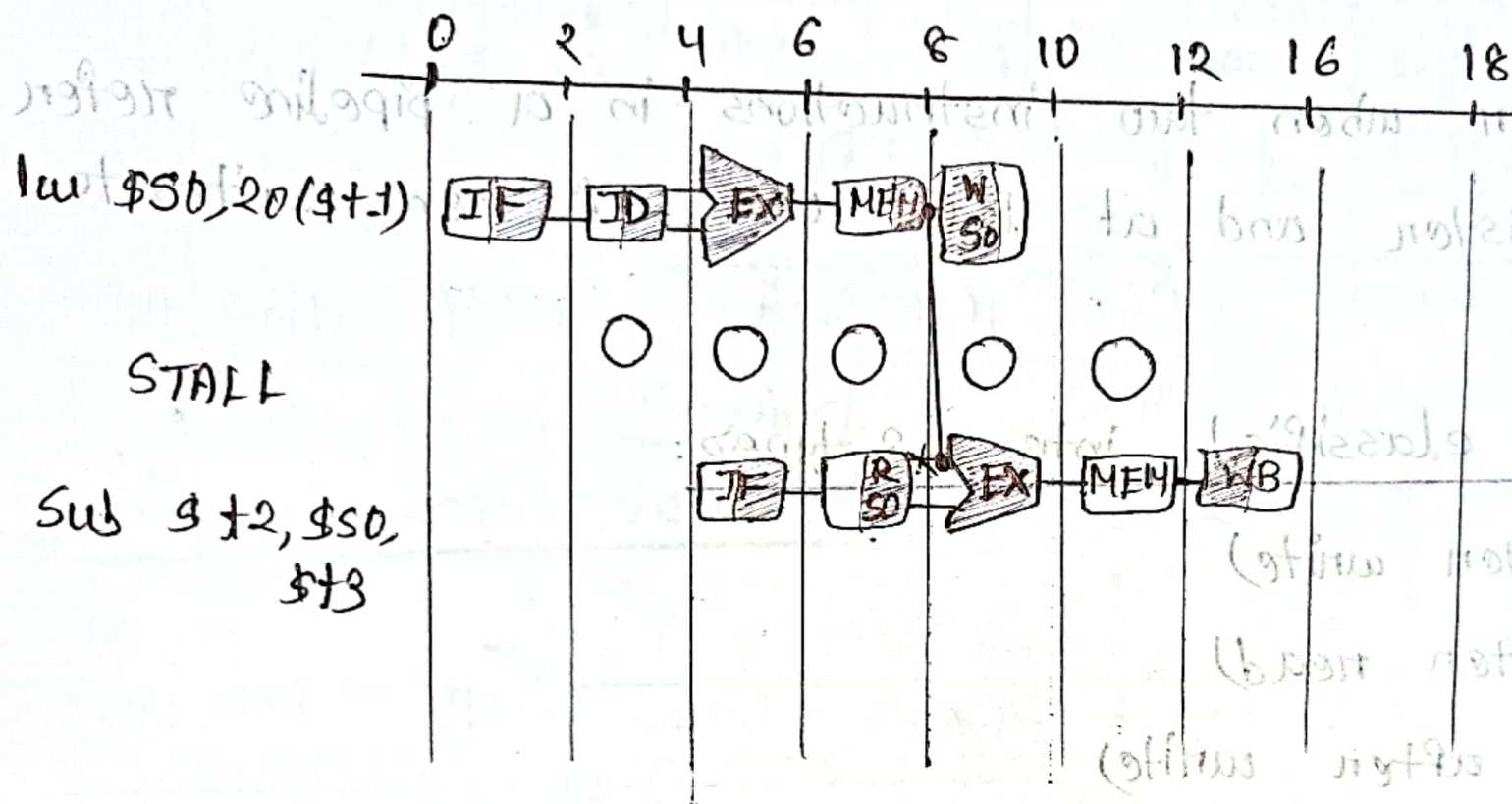
Data hazard are classified into 3 types:

- ① RAW (Read after write)
- ② WAR (Write after read)
- ③ WAW (write after write)

Solution for data hazard:

- ① Stalling
- ② Forwarding - Convert new value directly to next stage
- ③ Reordering

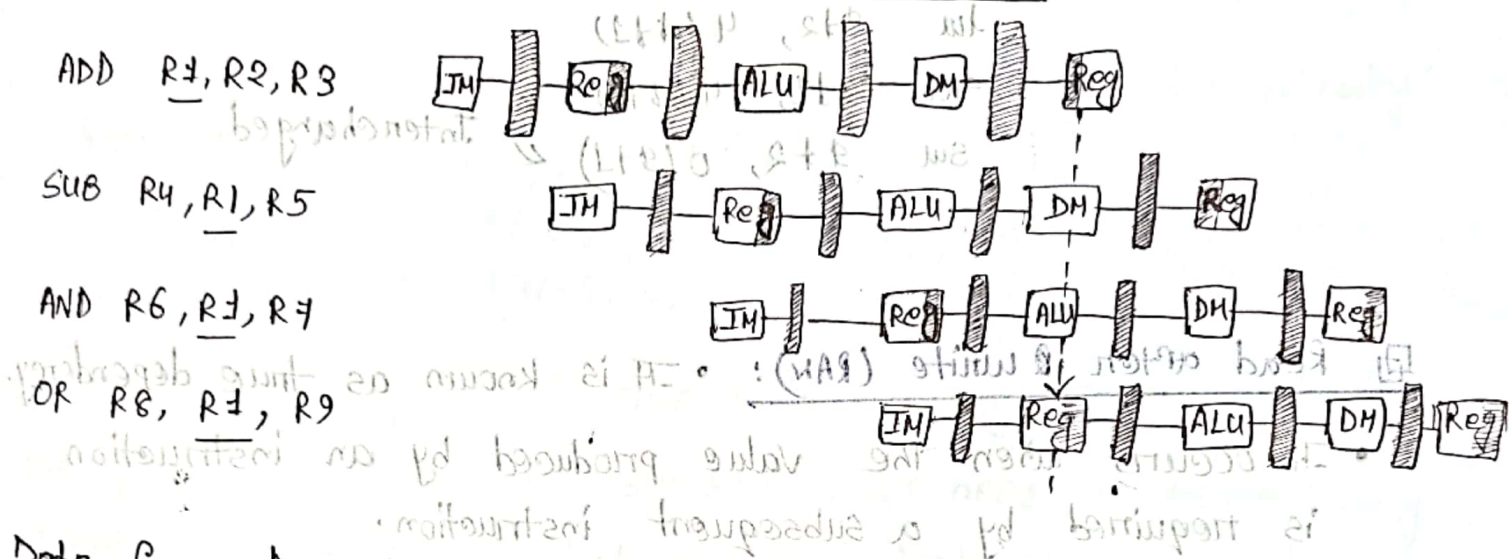
Stalling: stall still required for load - data avail. after



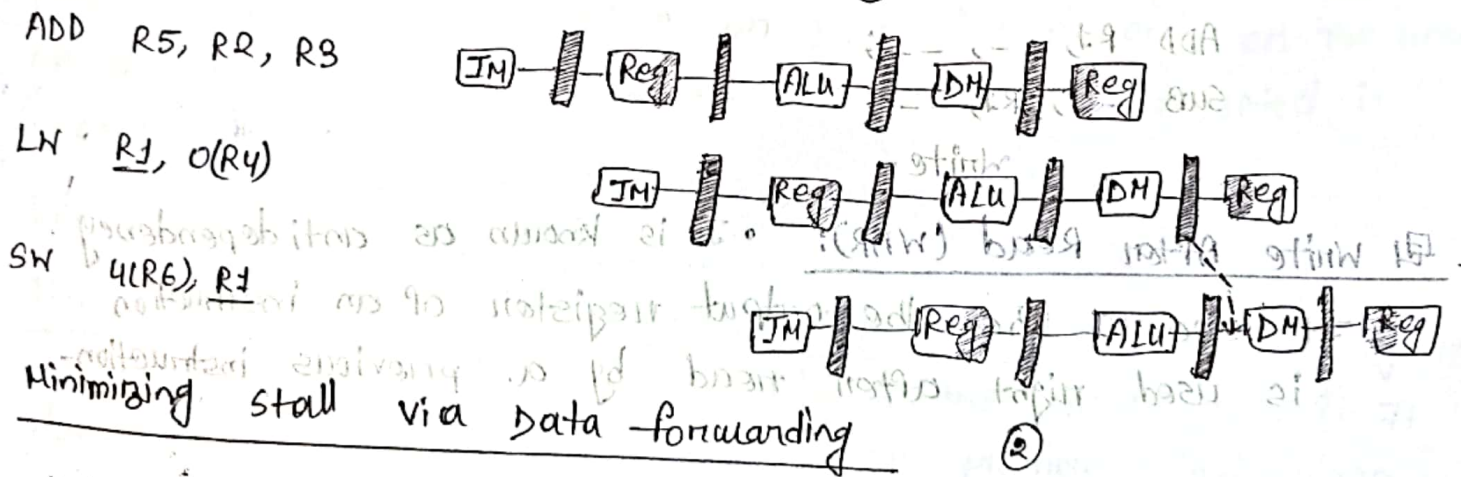
LW R1, 0(R2)	IF	ID	EX	MEM	WB		
SUB R4, R1, R5		IF	ID	EX	MEM	WB	
AND R6, R1, R7			IF	ID	EX	MEM	WB
OR R8, R1, R9				IF	ID	EX	MEM WB

LW R1, 0(R2)	IF	ID	EX	MEM	WB				
SUB R4, R1, R5		IF	ID	Stall	EX	MEM	WB		
AND R6, R1, R7			IF	Stall	ID	EX	MEM	WB	
OR R8, R1, R9				Stall	IF	ID	EX	MEM	WB

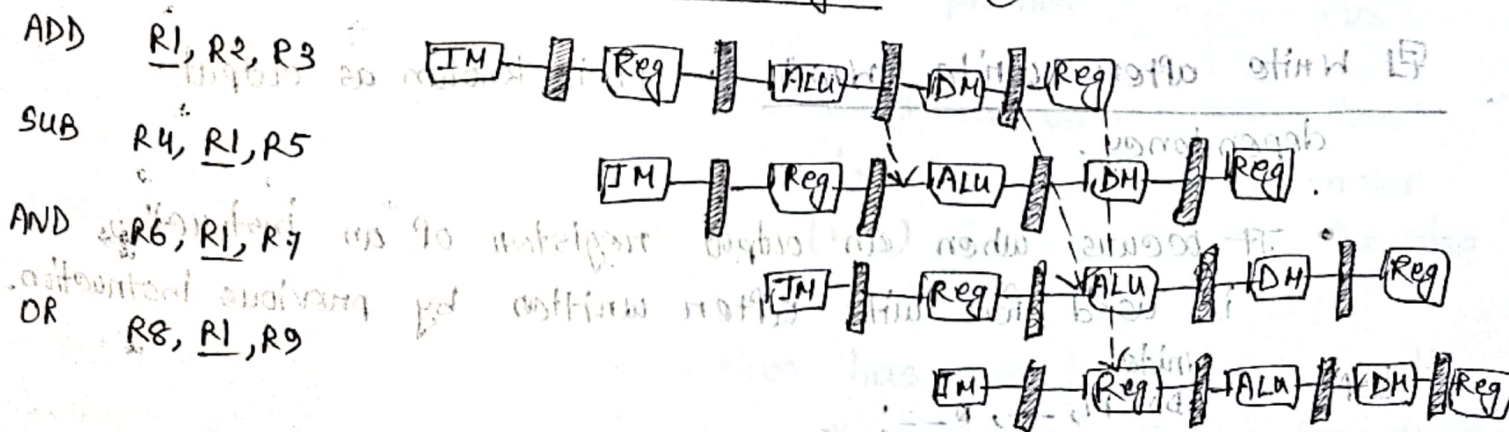
Forwarding: Register file: Reads after writes (1)



Data forwarding to MEM stage (3)



Minimizing stall via data forwarding (2)



Recording:

Exp:

	lw	sw	addw	sllw	srlw	mulw	divw	remw
lw	\$+0, 0 (\$+1)							
sw	\$+2, 4 (\$+1)							
addw	\$+2, 0 (\$+1)							
sllw	\$+0, 4 (\$+1)							

Data hazard

Recorded -

1m $5+0, 0(5+1)$

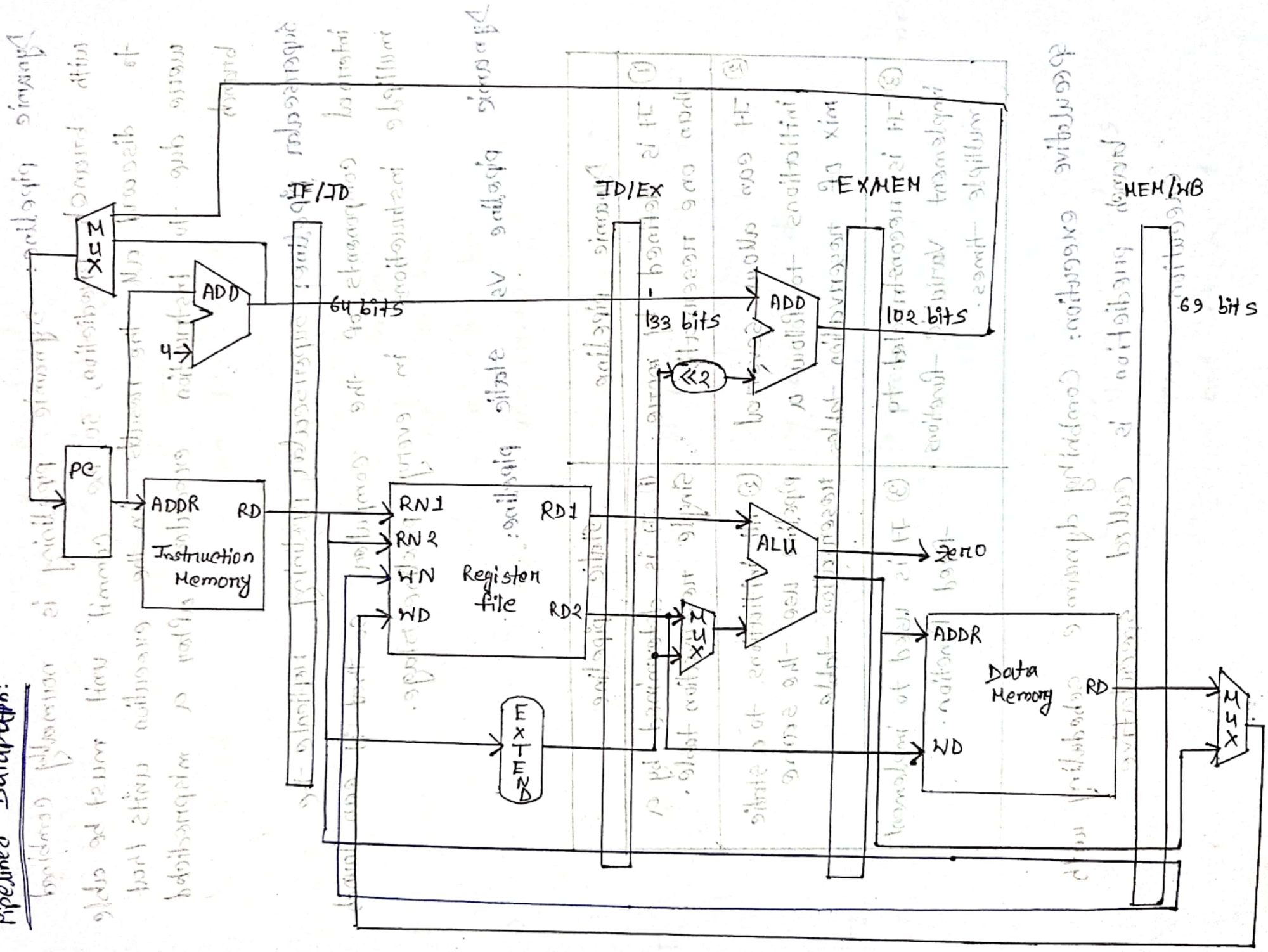
1m $5+2, 4(5+1)$

Sum $5+0, 4(5+1)$

Sum $5+2, 0(5+1)$

Intenchanged

Pipelined Datapath:



Dynamic pipeline: Dynamic pipelining is normally combined with branch prediction, so the commit unit must be able to discard all the results in the execution units that were due to instruction executed after a mispredicted branch.

Superscalar pipeline: Superscalar pipelining replicates the internal components of the computer so that it can launch multiple instructions in every pipeline stage.

Dynamic pipeline vs static pipeline:

Dynamic pipeline	Static pipeline
① It is defined by more than one reservation table.	① It is determined by a single reservation table.
② It can allow several initiations to follow a mix of reservation tables.	② All initiations to a static pipeline use the same reservation table.
③ It is reconstructed to implement variable functions multiple times.	③ It is used to implement fixed function.

Speculative execution: Combining dynamic scheduling with branch prediction is called speculative execution.